



Week 6 Solution

Question 1

Question

If the value of a current counting semaphore $s = -5$, what does this imply?

1. The number of processes that can enter the critical section is 2
2. The number of processes blocked from entering the critical section is 5
3. Total number of processes that can enter the critical section and remain in the blocked state is 5
4. None of the above

Answer

The number of processes blocked from entering the critical section is 5.

Solution

A **counting semaphore** can have a positive, zero, or negative value.

When the semaphore value becomes negative, its magnitude represents the number of processes waiting or blocked.

Therefore:

$$s = -5$$

means:

$$\text{Number of blocked processes} = 5$$

Hence, the correct answer is:

The number of processes blocked from entering the critical section is 5.

Question 2

Question

Which of the following conditions could occur due to lack of synchronization?

1. Inconsistency
2. Loss of information
3. Deadlock
4. All of these

Answer

All of these

Solution

Synchronization is used when multiple processes or threads access shared resources.

Without proper synchronization, several problems can occur.

Inconsistency

Two processes may access and modify shared data simultaneously, producing an inconsistent result.

Loss of information

One process may overwrite changes made by another process.

For example:

Initial value = 10

Process P1 reads 10

Process P2 reads 10

P1 writes 11

P2 writes 11

Although two increment operations occurred, the final value is only 11 instead of 12 .

Deadlock

Improper coordination between processes and resources can also lead to processes waiting indefinitely.

Therefore, the correct answer is:

All of these.

Question 3

Question

State True or False:

If the value of a counting semaphore $s = 4$, then the maximum number of requests for the critical section before it blocks is 0 .

Answer

False

Solution

If:

$s = 4$

the semaphore has four available units.

Therefore, up to four **P** or wait operations can proceed before the semaphore reaches zero and subsequent requests may need to block.

Conceptually:

Initial: $s = 4$

Request 1 $\rightarrow s = 3$

Request 2 $\rightarrow s = 2$

Request 3 $\rightarrow s = 1$

Request 4 $\rightarrow s = 0$

Next request $\rightarrow$ may block

Therefore, saying that the maximum number of requests before blocking is **0** is incorrect.

Correct Answer: False

Question 4

Question

A critical section is a program segment:

1. Which should run in a certain specified amount of time
2. Which should avoid deadlock
3. Which must be enclosed by a pair of semaphores, **P** and **V**
4. Where shared resources are accessed

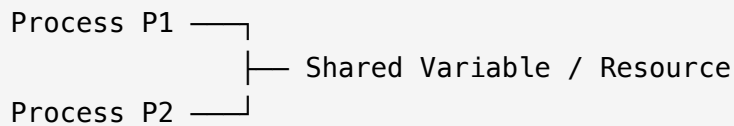
Answer

Where shared resources are accessed

Solution

A **critical section** is a part of a program where shared data or resources are accessed.

For example:



The code responsible for accessing this shared resource is the critical section.

For example:

```
counter = counter + 1;
```

If multiple processes execute this operation concurrently, the result may be incorrect.

Therefore, synchronization mechanisms such as semaphores or locks can be used to protect the critical section.

Hence:

Correct Answer: Where shared resources are accessed.

Question 5

Question

State True or False:

Peterson's solution is a synchronization mechanism that works for a maximum of 2 processes.

Answer

True

Solution

Peterson's solution is a classical software-based synchronization algorithm designed for **two processes**.

It uses shared variables such as:

- `flag[0]`
- `flag[1]`
- `turn`

The two processes use these variables to coordinate access to the critical section.

Conceptually:



The standard Peterson's algorithm is designed specifically for two processes.

Therefore:

Correct Answer: True

Question 6

Question

Consider two functions `P1` and `P2` running concurrently and updating two shared variables `A` and `B`, initially:

```
A = 1
B = 1
```

```
P1()
{
    A = A + 1;
    B = B + 1;
}

P2()
{
    A = A * 2;
    B = B * 2;
}
```

Assume that each statement is an atomic operation. What is the number of possible sets of values that `A` and `B` can take?

Answer

3

Solution

The operations are:

P1:

$A = A + 1$

$B = B + 1$

P2:

$A = A \times 2$

$B = B \times 2$

Initially:

$A = 1$

$B = 1$

The statements inside each process maintain their order, but the operations of **P1** and **P2** can interleave.

For each variable, two possible results exist depending on which operation occurs first:

If P1 executes before P2

1

↓ +1

2

↓ ×2

4

Result:

$A = 4$

Similarly:

$B = 4$

If P2 executes before P1

```
1  
↓ ×2  
2  
↓ +1  
3
```

Result:

A = 3

Similarly:

B = 3

Because the operations on **A** and **B** can interleave differently while preserving the order inside each process, the possible sets are:

```
(A, B) = (4, 4)  
(A, B) = (4, 3)  
(A, B) = (3, 4)
```

Therefore, the number of possible sets is:

3

Question 7

Question

The updated values **B** can possibly take, other than the initial value, are:

1. 1, 2, 3, 4
2. 2, 3
3. 1, 3, 4
4. 3, 4

Answer

3, 4

Solution

Initially:

$B = 1$

There are two operations on B :

P1: $B = B + 1$

P2: $B = B \times 2$

Case 1: P1 executes first

$B = 1$

↓ +1

$B = 2$

↓ ×2

$B = 4$

Final value:

$B = 4$

Case 2: P2 executes first

```
B = 1
↓ ×2
B = 2
↓ +1
B = 3
```

Final value:

```
B = 3
```

Therefore, the possible updated values are:

```
3, 4
```

Correct Answer: 3, 4

Question 8

Question

State True or False:

Xchng instruction returns 1 if the operation succeeds and returns 0 if the operation fails.

Answer

False

Solution

The XCHG instruction is an **exchange instruction**.

It exchanges the contents of two operands atomically in the appropriate context.

Conceptually:

Before:

A = 10

B = 20

XCHG A, B

After:

A = 20

B = 10

The purpose of `XCHG` is to exchange values. It is not generally defined as an instruction that returns:

1 → success

0 → failure

Therefore, the statement is:

False

Question 9

Question

Certain applications use counting semaphore operations in the following order:

10P, 2P, 3V, 2V, 4P, 3V

The final semaphore value is `-5`.

What is the initial value of the semaphore?

Answer

3

Solution

For a counting semaphore:

- P decreases the semaphore value.
- V increases the semaphore value.

Total P operations:

$$10P + 2P + 4P = 16P$$

Total V operations:

$$3V + 2V + 3V = 8V$$

Therefore, the net change is:

$$-16 + 8 = -8$$

Let the initial semaphore value be x .

The final value is given as:

$$x - 8 = -5$$

Therefore:

$$x = 3$$

Hence:

Correct Answer: 3

Question 10

Question

Consider the following statements:

```
i) if(count == 0)
ii) add %eax, %ebx
```

Which of the following is correct?

1. i is atomic, ii is not-atomic
2. i is not-atomic, ii is not-atomic
3. i is not-atomic, ii is atomic
4. Both are atomic

Answer

i is not-atomic, ii is atomic

Solution

Statement i

```
if(count == 0)
```

This high-level statement requires reading the value of `count` and then making a comparison.

At the program-operation level, it is not treated as one indivisible operation.

Therefore:

i → Not atomic

Statement ii

```
add %eax, %ebx
```

This is a single assembly instruction.

According to the assumption used in this question, this instruction is treated as an atomic operation.

Therefore:

ii → Atomic

Hence:

Correct Answer: i is not-atomic, ii is atomic.